

ERP System Development Analysis Report

Introduction

This report provides an analysis of the current state of the

ERP system development, based on the provided 'ProgressReportJune272025.docx' document. The goal is to provide comprehensive guidance for building a bespoke timber roof ERP system.

Current Development Progress

The Millennium Online Application is an ongoing development project aimed at creating a comprehensive ERP system. The system is being developed using .NET and C# and is hosted on Microsoft Azure, utilizing Azure SQL Server and SQL Database. A custom domain,

<https://cloud-mroofing.co.za>, has been set up for the Millennium Portal, secured with a valid SSL Certificate.

Implemented Features:

- **Stock Maintenance:** Basic inventory information, stock levels, re-order points, lead times, cost/selling prices, item types, UOM, UOM conversions, product properties, variants, attributes, markup/GP groups, discount groups, and print groups. Stock import from Excel is complete and JSON-based.
- **System Data Maintenance:** Users, customers, contacts, employees, contractors, and suppliers.
- **Project Import:** Ability to import Excel and JSOMN projects from PAMIR Design, with versioning for re-imports.
- **Email Engine:** For email communications.
- **PDF Export Engine:** For generating PDF documents (e.g., quotes).

- **Manual Quoting Engine:** In progress, with basic quote information, site details, quote line items, sales information, design details, notes, and addresses. Quote revisioning (A, B, C...) is implemented.
- **User Management:** Login controlled by a SQL Server 'Users' table with hashed passwords and role-based access (Admin, Read Only, Sales, Designer).
- **Dashboard:** Provides shortcuts to functions and quick insights.
- **Navigation:** Drill-down functionality from customer, contact, and enquiry records to linked entities.

Database Overview:

The database is structured into domains: Raw Excel Imports, Imports/Images, Design Data, Stock/Config, Quotes/Projects, and Core/Admin. Numerous tables have been implemented, including AppUsers , Contacts , JobCards , Tenders , Customers , Enquiries , Orders , QuoteVersions , QuoteLineItems , StockItems , and various prod_ and ref_ tables for product and reference data, as well as xls_ tables for imported Excel data.

Key Status Summary:

- **Stock Import:** Complete (JSON-based).
- **Product Properties:** Complete (UI and DB linked).
- **Manual Quoting:** In Progress (calculations being added).
- **Orders / Tenders:** Planned (after quoting is stable).
- **Quote Revisioning:** Done (Auto revision A, B, C...).
- **Project Import (Excel):** Complete (Live-tested with versioning).

Identified Gaps and Challenges

Based on the user's initial request and the provided progress report, several key areas require further attention and development:

1. **Mitek Pamir Integration (Data Consumption):** The user explicitly stated the need for the system to "consume data exported from the local Mitek Pamir instance (Variables etc.)". While the report mentions "Import Excel Project" and "Import JSOMN Projects" from PAMIR Design, it doesn't detail how "Variables" from Pamir are consumed and utilized for quotation formulas. The current import seems to be more about project data rather than specific variables for calculations.
2. **Quotation Formulas and Stock Item Integration:** The user wants quotations to be "made up of stock items with formula's that read the variables and populate a qty." The report states "Manual Quoting is in progress" and "Calculations being added," but the specifics of how these formulas will read Pamir variables and dynamically populate quantities are not detailed. The current stock maintenance focuses on general inventory info, not dynamic quantity calculation based on external variables.
3. **Robust Stock Module:** The user emphasized that "The stock module needs to be very robust to handle the different intricacies we have with stock." While the report lists many stock-related attributes (Item Types, UOM, Conversions, Product Properties, Assemblies & Components, Alternatives, Formula, Variants, Attributes, Markup/GP Group, Discount Group, Print Group), the detailed intricacies of timber roof stock (e.g., specific dimensions, grades, treatments, or how these affect pricing and quantity calculations) are not fully elaborated.
4. **Assemblies and BOMs (Bill of Materials) Functionality:** The report distinguishes between "Assemblies" (for pricing tenders, fixed composition) and "BOMs" (for production orders, variable composition). The user's request for "control the production of the trusses" directly relates to BOM functionality. The report states this needs to be done at a "quote, tender or order level" and that a "production item in stock (No Components)" will then request a list of components specific to that order. This implies a need for a dynamic BOM generation system based on the specific roof design and Pamir variables.
5. **Order Management and Workflow Automation:** The user's vision includes creating an order from an approved quote, triggering a credit approval process, and controlling production, delivery, and installation. The report explicitly states "Orders / Tenders" are "Planned" after quoting is stable. This is a significant gap that needs detailed planning.

6. **Financial Management (General Ledger, Bank Recons, Purchases):** The user requires the system to "do all the financial entries in a general ledger and also have other financial functionality, bank recons, purchases etc etc." The current report does not mention any financial modules beyond basic cost/selling prices in stock items. This is a major missing component.
7. **Integration with Microsoft Dynamics 365:** The user currently uses Dynamics 365 to duplicate information and generate quotes for version control. The new system aims to replace or integrate with this. The report doesn't mention any plans for integration with Dynamics 365, which is crucial for a smooth transition or co-existence.
8. **User Experience for Quoting:** The user provided detailed feedback on the quoting process, specifically the need for a blank grid with real-time filtering for stock items, template import, and clear option/revision management. While the report acknowledges "Manual Quoting is in progress" and "Quote Revisioning is done," the specific UI/UX elements requested by the user need to be addressed in the design.

These challenges highlight the need for a robust architecture that can handle complex data integration, dynamic calculations, and comprehensive workflow management, extending beyond the current scope of the implemented features.

Research Findings: Mitek Pamir Integration Challenges

Initial research into Mitek Pamir data export capabilities and integration methods has proven challenging. Direct searches for

"Mitek Pamir API" or "Mitek Pamir database schema" did not yield specific technical documentation or readily available APIs for direct integration. The official MiTek customer experience portal and product pages primarily highlight the software's design and output capabilities (PDF, DXF, DWG, IFC export) rather than detailed data export formats for variables or direct database access.

Attempts to navigate MiTek's websites directly also encountered issues, including SSL handshake failures, which further hindered direct information gathering. This suggests that direct programmatic access to Mitek Pamir's internal data, especially the variables used in design for quotation calculations, is not publicly documented or easily accessible.

Given the user's current process of exporting data from Pamir to Excel and then duplicating it in Dynamics 365, it is highly probable that the "variables" and design parameters needed for quotation formulas are embedded within these Excel or JSOMN export files. The challenge lies in precisely identifying and extracting these variables from the exported files in a structured and reliable manner.

Therefore, the next steps in addressing the Mitek Pamir integration will focus on:

1. **Detailed analysis of Pamir Excel/JSOMN export files:** Since direct API or database schema information is unavailable, the most viable approach is to thoroughly analyze the structure and content of the Excel and JSOMN files exported from Mitek Pamir. This will involve understanding how design variables, material quantities, and other relevant data are represented within these files.
2. **Developing a robust parsing mechanism:** Based on the analysis of the export files, a custom parsing mechanism (e.g., using .NET/C# given the current development stack) will need to be developed to accurately extract the required variables and data points.
3. **Mapping extracted data to ERP stock items and formulas:** Once the data is extracted, it needs to be mapped to the ERP system's stock items and used within the quotation formulas to dynamically populate quantities and costs.

This approach acknowledges the limitations of direct integration and leverages the existing export functionality of Mitek Pamir, which the user is already utilizing. It shifts the focus from direct integration to intelligent data extraction and transformation from the available export formats.

Research Findings: CAD/BIM to ERP Integration Best Practices

Integrating CAD/BIM (Computer-Aided Design/Building Information Modeling) data with ERP (Enterprise Resource Planning) systems is a critical aspect of modern construction and manufacturing. While direct integration with Mitek Pamir is not straightforward, understanding general best practices for CAD/BIM to ERP integration provides a valuable framework for designing the bespoke ERP system.

Key best practices and considerations identified from research include:

1. **Understanding Data Flow and Mapping:** A fundamental step is to clearly define how data will flow between the design software (Mitek Pamir, via its exports) and the ERP system. This involves meticulous data mapping—identifying which specific data points from Pamir exports (e.g., dimensions, material types, quantities, design parameters) correspond to fields and attributes within the ERP system, particularly for stock items, BOMs, and quotation formulas. The goal is to ensure that design intent is accurately translated into actionable ERP data.
2. **Choosing the Right Integration Method:** Various methods exist for integrating disparate systems. Given the challenges with direct API access to Mitek Pamir, the current approach of leveraging Excel/JSON exports necessitates a robust data extraction and transformation (ETL) process. This could involve:
 - **File-based Integration:** As currently envisioned, parsing exported files (Excel, JSON) from Pamir. This requires custom development for parsing and validation.
 - **Database-level Integration (less likely for Pamir):** If direct access to Pamir's local SQL database were feasible and permissible, this would offer a more direct integration point, but it comes with complexities related to schema understanding and potential vendor support issues.
 - **Middleware/Integration Platforms:** For more complex scenarios, using integration platforms (iPaaS) could streamline data exchange, but this might be an overkill for the initial phase given the file-based export.
3. **Ensuring Data Integrity and Quality:** Maintaining data consistency and accuracy across both systems is paramount. Any data extracted from Pamir exports must be validated before being imported into the ERP. This minimizes errors in quotations, production, and financial records. Establishing clear data ownership and validation rules is crucial.
4. **Automating Workflows:** The ultimate goal of integration is to automate manual processes. For the timber roof ERP, this means automating the creation of quotations and BOMs based on Pamir design data. This reduces manual data entry, speeds up the quoting process, and minimizes human error. Automation should extend to triggering

subsequent processes like credit approval and production planning once a quote is approved.

5. **Robust Bill of Materials (BOM) Management:** CAD/BIM data is a rich source for generating accurate BOMs. The ERP system needs to effectively consume this data to create detailed BOMs for production. This includes handling both fixed assemblies and variable BOMs (as described for roof trusses), where quantities and components change based on design parameters. The system should be able to dynamically calculate component quantities based on the extracted Pamir variables.
6. **Scalability and Flexibility:** The integration solution should be designed to be scalable and flexible enough to accommodate future changes in Pamir export formats (if they occur) or evolving business requirements. This suggests a modular approach to the data extraction and mapping components.
7. **User Experience and Training:** While technical integration is key, the user experience for those interacting with the integrated system is equally important. The ERP system should present the Pamir data in an intuitive way, allowing users to easily review, modify, and approve quotations and production plans. Adequate training will be necessary for users to effectively leverage the integrated system.

These best practices will guide the architectural design of the bespoke ERP system, particularly in how it ingests and processes data from Mitek Pamir exports to drive core ERP functionalities like quoting, BOM generation, and production planning.

Research Findings: Robust Stock Module Design for Timber Products

Designing a robust stock module for a timber roof business requires addressing the unique characteristics of timber products, which go beyond typical inventory management. The research highlights several key features and considerations for such a module:

1. **Granular Product Definition and Attributes:** Timber products are highly variable. A robust stock module must support detailed attributes beyond basic item codes. This includes:

- **Dimensions:** Length, width, thickness (e.g., 38x114x5.4 timber). The system should allow for tracking and managing stock based on these specific dimensions.
- **Grades:** Timber often comes in different structural or visual grades, which affect its application and pricing. The system needs to accommodate these distinctions.
- **Species:** Different wood species have varying properties and costs.
- **Treatments:** Information on any chemical treatments (e.g., fire retardant, preservative) applied to the timber.
- **Other Properties:** Moisture content, kiln-drying status, and other relevant physical properties.

The existing system already includes

Product Properties, UOM, and UOM Conversions, which is a good foundation, but it needs to be ensured that these are sufficiently granular for timber-specific attributes.

1. **Flexible Units of Measure (UOM) and Conversions:** Timber is often bought, sold, and used in different units (e.g., linear meters, cubic meters, square meters, pieces). The system must handle complex UOM conversions seamlessly. For example, converting a quantity of timber in linear meters to cubic meters based on its cross-sectional dimensions, or calculating the number of pieces from a total length.
2. **Real-time Inventory Tracking and Visibility:** Accurate, real-time visibility into stock levels is crucial to prevent stockouts or overstocking. This includes tracking inventory across different locations (e.g., warehouse, production floor, on-site). Features like automatic depletion of inventory upon sale or use in production are essential.
3. **Batch and Lot Tracking:** For quality control and traceability, especially in the event of recalls or issues with specific timber batches, the ability to track inventory by batch or lot number is highly beneficial.
4. **Dynamic Pricing and Costing:** The system should support various costing methods (e.g., FIFO, LIFO, average cost) and allow for dynamic pricing based on attributes like grade, species, and current market conditions. The ability to apply markups and discounts based on customer groups or product categories is also important.

5. **Integration with Production and Sales:** The stock module should be tightly integrated with production planning (BOMs) and sales (quotations and orders). This ensures that raw material availability is considered during quoting and that finished goods are accurately tracked as they move through the production process and out for delivery.
6. **Demand Forecasting and Reorder Points:** To optimize inventory levels, the system should support demand forecasting based on historical sales data and automatically trigger reorder alerts when stock falls below predefined reorder points.
7. **Assembly and Kit Management:** As identified in the current system, the ability to manage both fixed assemblies (for tenders) and variable BOMs (for roof trusses) is critical. The stock module needs to support the definition of components for assemblies and provide mechanisms for dynamic BOM generation based on design parameters.
8. **Reporting and Analytics:** Comprehensive reporting capabilities are needed to analyze inventory turnover, carrying costs, stock aging, and other key performance indicators relevant to timber inventory management.

By incorporating these features, the bespoke ERP system can provide the detailed control and visibility required for managing timber roof product inventory effectively, supporting accurate quoting, efficient production, and improved financial management.

Research Findings: Advanced BOM and Assembly Management

Effective management of Bills of Materials (BOMs) and assemblies is central to the bespoke ERP system, especially given the variable nature of timber roof trusses. The research highlights the importance of advanced BOM capabilities for configurable and custom-manufactured products:

1. **Configurable BOMs (CBOMs):** For products with variations, such as different roof truss designs, CBOMs are essential. Instead of creating a new BOM for every possible variation, a CBOM defines a base product and then allows for the selection of specific options or features that dynamically alter the BOM. This is particularly relevant for the

user's need to handle both fixed assemblies (for tenders) and variable BOMs (for production orders).

2. **Dynamic BOM Generation:** The ERP system needs the capability to generate BOMs dynamically based on design parameters and variables. This directly addresses the user's requirement for BOMs that change with each roof, where components are determined by the specific order. This dynamic generation can be driven by data extracted from Mitek Pamir exports.
3. **Product Configurators:** A product configurator is a powerful tool that can be integrated with the ERP system to automate the creation of quotations and BOMs for made-to-order products. It allows users to define product options, features, and rules, which then automatically generate the correct BOM, routing, and pricing. For the timber roof ERP, a configurator could take inputs from the Pamir data (or user selections based on Pamir data) and generate the precise list of timber components, connectors, and other items required for a specific truss design.
4. **Multi-level BOMs:** Complex products often require multi-level BOMs, where sub-assemblies are nested within a higher-level assembly. For roof trusses, this could mean a top-level BOM for the entire roof structure, with sub-BOMs for individual trusses, and further levels for the components within each truss.
5. **Integration with Quoting and Production:** The BOM management system must be tightly integrated with the quoting engine to ensure that accurate material costs are reflected in quotations. Once a quote is approved and converted to an order, the dynamic BOM should seamlessly feed into production planning, material procurement, and inventory management.
6. **Version Control and Revision Management:** As the user highlighted with quote revisions, BOMs also require robust version control. Any changes to a design or component should trigger a new BOM revision, ensuring traceability and preventing errors in production.

By implementing these advanced BOM and assembly management features, the bespoke ERP system can effectively translate complex timber roof designs into accurate material requirements, streamline the quoting process, and optimize production planning.

Research Findings: Financial Module Integration

The user explicitly requested comprehensive financial functionality within the bespoke ERP system, including general ledger (GL), bank reconciliations, and purchase management. This research confirms that these are core components of any robust ERP system and outlines key features and best practices for their integration:

1. General Ledger (GL):

- **Centralized Financial Data:** The GL is the heart of the ERP's financial module, serving as the central repository for all financial transactions. It aggregates data from all other modules (sales, purchasing, inventory, production) to provide a complete financial picture.
- **Chart of Accounts:** A well-defined and organized chart of accounts is crucial for accurate financial reporting. Best practices suggest a clear structure and consistent naming conventions.
- **Automated Entries:** The ERP should automate the posting of financial entries from various business processes (e.g., sales orders, purchase orders, inventory movements) directly to the GL, reducing manual effort and errors.
- **Real-time Reporting:** The GL should support real-time financial reporting, including profit and loss statements, balance sheets, and cash flow statements, providing immediate insights into the company's financial health.

2. Bank Reconciliation:

- **Automation:** Modern ERP systems offer automated bank reconciliation features that match bank statement transactions with internal ledger entries. This significantly reduces the time and effort involved in manual reconciliation.
- **Configurable Matching Rules:** The system should allow for the configuration of matching rules to handle various transaction types, including fees, interest, and foreign exchange differences.
- **Multi-Bank Integration:** Ability to connect and reconcile multiple bank accounts seamlessly.

- **Exception Handling:** Tools for easily identifying and resolving discrepancies between bank statements and internal records.

3. Purchase Management (Procurement):

- **Purchase Order (PO) Management:** The system should facilitate the creation, approval, and tracking of purchase orders from requisition to receipt of goods and invoice matching. This ensures proper authorization and control over expenditures.
- **Supplier Management:** Centralized management of supplier information, including contracts, pricing, and performance.
- **Automated Workflows:** Automation of procurement processes, such as generating purchase requisitions based on inventory levels or production needs, sending POs to suppliers, and receiving goods.
- **Integration with Inventory and Accounts Payable:** Seamless integration with the inventory module to update stock levels upon receipt of goods and with accounts payable for invoice processing and payment.
- **Cost Control:** Features for tracking and analyzing purchasing costs, identifying cost-saving opportunities, and managing vendor relationships effectively.

Integrating these financial modules will provide the user with comprehensive control over their financial operations, enabling accurate record-keeping, efficient reconciliation, and strategic purchasing decisions. The fact that the current development is on Microsoft Azure and uses SQL Server aligns well with the capabilities of many modern ERP financial modules, including those found in Dynamics 365, which the user is already familiar with.

Research Findings: Dynamics 365 Integration Patterns

The user currently leverages Dynamics 365 for quote version control, indicating a need to either migrate from or integrate with this existing system. Research into Dynamics 365 integration patterns provides various strategies for handling this transition or co-existence:

1. **Integration Scenarios:** Dynamics 365 supports a range of integration scenarios, from simple data exchange to complex workflow orchestration. The choice of pattern

depends on the specific business requirements, data volume, and real-time needs.

2. Common Integration Patterns:

- **Direct Integration (API-based):** Dynamics 365 exposes various APIs (e.g., OData, custom services) that allow external systems to interact with its data and functionalities. This enables real-time or near real-time data synchronization. If the bespoke ERP needs to push or pull data directly from Dynamics 365 (e.g., for customer information, historical quotes), this would be a primary method.
- **File-based Integration:** While less real-time, file-based integration (e.g., CSV, XML) can be used for bulk data transfers, such as migrating historical quote data or periodic synchronization of master data. The user's current process of exporting from Pamir and duplicating in Dynamics 365 suggests familiarity with file-based approaches.
- **Middleware/Integration Platforms:** For more complex integrations involving multiple systems or intricate business logic, middleware solutions (e.g., Microsoft Azure Integration Services like Logic Apps, Azure Service Bus, or third-party integration platforms) can act as a central hub for data transformation and routing.

3. Data Migration Best Practices: If the bespoke ERP system is intended to fully replace Dynamics 365 for quoting and related processes, a well-planned data migration is crucial. Key considerations include:

- **Data Mapping:** Meticulously mapping data fields between Dynamics 365 and the new ERP system to ensure data integrity and consistency.
- **Data Quality:** Cleaning and validating data before migration to avoid transferring erroneous or incomplete information.
- **Incremental Migration:** For large datasets, migrating data in phases or increments to minimize disruption and allow for thorough testing.
- **Validation and Testing:** Rigorous testing post-migration to ensure all data has been accurately transferred and business processes function as expected.

4. Co-existence Strategy: If Dynamics 365 will continue to be used for certain functionalities (e.g., financial reporting, CRM), the bespoke ERP system will need to

integrate seamlessly with it. This would involve defining clear boundaries of responsibility for each system and establishing robust data synchronization mechanisms.

Given that the new ERP system is being developed on Microsoft Azure, leveraging Azure-native integration services would be a natural fit for connecting with Dynamics 365, whether for data migration or ongoing co-existence. The goal should be to minimize manual data duplication and ensure a single source of truth for critical business information.

Proposed System Architecture and Data Flow

This section outlines the proposed system architecture and data flow for the bespoke ERP system, designed to address the identified gaps and leverage the existing development efforts. The architecture emphasizes modularity, scalability, and seamless integration, particularly with Mitek Pamir and the existing Microsoft ecosystem.

Overall Architecture Overview

The bespoke ERP system will be a cloud-native application hosted on Microsoft Azure, aligning with the current development environment. It will follow a microservices-oriented architecture, allowing for independent development, deployment, and scaling of different functional modules. The core components will include:

1. **Data Ingestion Layer:** Responsible for receiving and processing data from external sources, primarily Mitek Pamir exports.
2. **Core ERP Modules:** Encompassing Stock Management, Quotation Management, Bill of Materials (BOM) & Production Planning, Order Management, and Financial Management.
3. **Integration Layer:** Facilitating communication with other systems, including potential co-existence or migration from Dynamics 365.
4. **User Interface (UI) Layer:** Providing intuitive web-based interfaces for various user roles.

5. **Reporting & Analytics Layer:** Enabling comprehensive insights and business intelligence.

Mitek Pamir Integration and Data Flow

Given the challenges with direct API access to Mitek Pamir, the integration strategy will focus on robustly consuming the existing Excel and JSOMN export files. This approach minimizes reliance on proprietary Pamir interfaces and leverages the user's current workflow.

Data Flow Steps:

1. **Pamir Export:** The design data from Mitek Pamir will continue to be exported by the user into Excel or JSOMN files. These files contain critical information about roof designs, including dimensions, material types, quantities, and other design parameters (referred to as "variables" by the user).
2. **File Upload/Ingestion:** The exported Excel/JSOMN files will be uploaded to the Azure-hosted ERP system. This can be facilitated through a dedicated file upload interface within the ERP application or via automated file transfer mechanisms (e.g., Azure Blob Storage with event triggers).
3. **Data Parsing and Extraction:** This is a critical component. A custom-built parsing engine (developed in .NET/C#) will read and interpret the structure of the Excel/JSOMN files. This engine will be responsible for:
 - **Schema Discovery:** Dynamically identifying the relevant sheets, tables, and columns within the Excel files, or parsing the JSON structure of JSOMN files.
 - **Variable Extraction:** Accurately extracting specific design variables and parameters (e.g., truss dimensions, pitch, span, material grades, connector types, quantities) that are essential for quotation calculations and BOM generation.
 - **Data Validation:** Performing initial validation checks on the extracted data to ensure its integrity and consistency before further processing.
4. **Data Transformation and Normalization:** The extracted raw data from Pamir exports will likely be in a format optimized for Pamir itself, not necessarily for the ERP system.

This step involves transforming and normalizing the data into a standardized format that aligns with the ERP's internal data model. This includes:

- **Unit Conversion:** Converting units of measure as necessary (e.g., converting linear feet to meters, or pieces to square meters based on product specifications).
- **Data Type Conversion:** Ensuring that numerical values, dates, and text fields are correctly parsed and stored with appropriate data types.
- **Mapping to ERP Entities:** Mapping the extracted Pamir data to corresponding entities within the ERP system, such as `ProjectQuotes` , `QuoteLineItems` , and `StockItems` .

5. **Temporary Staging (Optional but Recommended):** For complex or large imports, a temporary staging area (e.g., a dedicated table in Azure SQL Database) can be used to hold the parsed and transformed data before it is committed to the main ERP tables. This allows for additional review, error correction, and reconciliation if needed.

6. **Integration with Core ERP Modules:** Once the Pamir data is parsed, extracted, transformed, and validated, it will be used to drive core ERP functionalities:

- **Quotation Generation:** The extracted design variables will feed into the quotation engine to dynamically calculate quantities of stock items and generate accurate pricing based on predefined formulas. This directly addresses the user's need for

quotations made up of stock items with formulas that read variables and populate quantities.

* **BOM Generation:** The design parameters will be used to dynamically generate or update Bill of Materials for roof trusses, ensuring that the correct components and quantities are specified for production.

This integration strategy provides a clear path for leveraging existing Pamir exports while building a robust and scalable ERP system on Azure. It acknowledges the limitations of direct Pamir integration and focuses on intelligent data processing to achieve the desired business outcomes.

Stock Management Module

The stock module will be designed to be highly robust, accommodating the intricacies of timber products. Building upon the existing foundation, it will incorporate features to handle granular product definitions, flexible units of measure, and dynamic costing.

Key Features and Data Flow:

- Enhanced Product Attributes:** Extend the existing `StockItems` table and related product property tables (`prod_ProductProperty` , `ref_ItemTypeProperty`) to store detailed timber-specific attributes such as:
 - **Dimensions:** Length, width, thickness (e.g., `length_mm` , `width_mm` , `thickness_mm`).
 - **Grade:** (e.g., `grade_type_id` linking to a `ref_GradeTypes` table).
 - **Species:** (e.g., `species_type_id` linking to a `ref_SpeciesTypes` table).
 - **Treatment:** (e.g., `treatment_type_id` linking to a `ref_TreatmentTypes` table).
 - **Surface Finish:** (e.g., `finish_type_id` linking to a `ref_FinishTypes` table).This allows for precise categorization and tracking of timber inventory.
- Flexible Unit of Measure (UOM) Management:** The existing `ref_UOM` and `ref_UOMConversion` tables are crucial. The system will support complex conversions, enabling calculations between linear meters, cubic meters, square meters, and individual pieces based on the product dimensions. This is vital for accurate purchasing, sales, and production planning.
- Real-time Inventory Tracking:** Implement mechanisms for real-time updates of stock levels across various locations (e.g., `warehouse_id` , `production_area_id` , `on_site_id`). This will involve:
 - **Automated Depletion:** Integration with the Production and Sales modules to automatically deduct stock upon consumption in production or shipment for sales orders.
 - **Receiving and Issuing:** Processes for recording incoming goods (purchases) and outgoing goods (issues to production or sales).

4. **Batch and Lot Tracking:** Introduce fields and processes to track timber by batch or lot numbers (`batch_number` , `lot_number`) for improved traceability and quality control. This is particularly important for timber, where variations within a species or grade might exist across different batches.
5. **Dynamic Pricing and Costing:** The `Cost Price` and `Selling Price` in `StockItems` will be dynamic, potentially influenced by:
 - **Product Attributes:** Different grades or species may have different base costs.
 - **Market Conditions:** Integration with external market data (if available) or manual updates to reflect current timber prices.
 - **Markup/Discount Groups:** Leverage the existing `Markup / GP Group` and `Discount Group` functionalities to apply pricing rules based on customer segments or order volumes.
6. **Integration with Pamir Data:** The extracted variables from Pamir exports will be used to determine the specific timber products required. For example, a Pamir export might specify a certain dimension and grade of timber, which the ERP system will then match to an existing `StockItem` or trigger a procurement process if not in stock.

Quotation Management Module

The quotation management module will be enhanced to leverage the detailed stock information and the data extracted from Mitek Pamir, enabling dynamic and accurate quote generation. The user's feedback on the quoting process, particularly the need for dynamic filtering and template-based entry, will be central to the UI/UX design.

Key Features and Data Flow:

1. **Dynamic Quote Line Item Generation:** Instead of manual entry of all components, the system will use the Pamir-extracted variables and predefined formulas to automatically populate `QuoteLineItems` . For example, if Pamir indicates a roof design requires X linear meters of a specific timber profile, the system will calculate the number of pieces or cubic meters based on the stock item's UOM conversions and add it to the quote.

2. **Formula Engine:** A robust formula engine will be developed to read the Pamir variables and apply complex calculations to determine quantities, costs, and selling prices for each line item. These formulas will be associated with `StockItems` or `ItemTypes` and can incorporate various factors such as waste percentages, labor costs, and transport costs.
3. **Real-time Stock Availability Check:** During quotation creation, the system will provide real-time feedback on the availability of required stock items. This can trigger alerts for low stock or initiate purchase requisitions for materials needed to fulfill the potential order.
4. **Template-based Quoting:** Support for predefined quote templates (as requested by the user) will streamline the process for common roof types or standard packages. These templates can pre-populate certain line items, which can then be adjusted based on Pamir data or user input.
5. **Interactive Quote Line Item Grid:** The UI for adding quote line items will feature a dynamic, blank grid with real-time filtering as the user types, allowing for quick selection of stock items. This addresses the user's need to avoid a full stock list and efficiently find specific items.
6. **Revision and Option Management:** The existing `QuoteVersions` and `QuoteHistories` tables will be fully utilized to manage multiple options and revisions for each enquiry. The system will clearly display the relationships between options and revisions, allowing users to compare and select the most appropriate quote.
7. **Approval Workflow:** Integrate a configurable approval workflow for quotes (e.g., `Draft` , `Send for Approval` , `Approved` , `Sent To Client`). This ensures that quotes are reviewed and authorized before being sent to customers.
8. **PDF Export and Email Engine:** Leverage the existing PDF Export Engine to generate professional-looking quotations and the Email Engine for direct communication with clients.

Bill of Materials (BOM) and Production Planning Module

This module is critical for translating design data into manufacturing instructions and managing the production of roof trusses. It will build upon the distinction between Assemblies (fixed composition) and BOMs (variable composition) identified in the current system.

Key Features and Data Flow:

1. **Dynamic BOM Generation for Trusses:** For roof trusses, the system will dynamically generate the BOM based on the specific design parameters extracted from Pamir exports. This means that for each unique truss design, the system will calculate the exact quantities of timber (by dimension, grade, etc.), nail plates, connectors, and other components required. This will involve:

- **Parametric BOMs:** Defining BOMs with parameters that are populated by the Pamir data. For example, a truss BOM might have a parameter for

the length of a chord, and the system will calculate the number of timber pieces needed based on this length.

* **Rules-based Configuration:** Implementing rules that determine which components are included and in what quantities based on the design variables (e.g., if pitch > X, add Y bracing).

1. **Assembly Management for Tenders:** The existing concept of

"Assemblies" for tenders (fixed composition) will be fully supported. These assemblies will have predefined components and quantities, primarily used for pricing and quoting purposes, as they represent standard supply and install packages.

1. **Production Order Creation:** Once a quote is approved and converted into an order, the system will automatically generate production orders based on the dynamically created BOMs for trusses and the predefined assemblies for other items. Each production order will detail the exact materials and labor required.
2. **Work Order Management:** The module will manage the lifecycle of production orders, including:
 - **Scheduling:** Integrating with a scheduling system to plan production activities and allocate resources.

- **Material Issuance:** Tracking the issuance of raw materials from stock to production.
 - **Progress Tracking:** Monitoring the progress of each production order through various stages (e.g., cutting, assembly, quality control).
 - **Labor Tracking:** Recording labor hours against production orders for costing and payroll purposes.
3. **Integration with Stock Module:** Seamless integration with the Stock Management module will ensure that:
- **Material Availability Check:** Production orders can only be initiated if all required materials are in stock or have been procured.
 - **Automatic Stock Depletion:** Materials are automatically deducted from inventory upon issuance to production orders.
 - **Finished Goods Receipt:** Finished trusses and assemblies are received into inventory upon completion of production.
4. **Quality Control and Rework:** Features to manage quality checks at various stages of production and handle any rework processes, linking back to material and labor consumption.

Order Management Module

The Order Management module will oversee the entire sales order lifecycle, from approved quotation to delivery and installation, integrating with production and financial processes.

Key Features and Data Flow:

1. **Order Creation from Approved Quotes:** The system will allow for the seamless conversion of an approved quotation into a sales order. This will automatically transfer all relevant information, including customer details, site information, line items, pricing, and associated BOMs.
2. **Credit Approval Workflow:** As requested by the user, a configurable credit approval process will be triggered upon order creation. This workflow will involve:

- **Automated Checks:** Integration with financial systems or external credit agencies for automated credit checks.
 - **Manual Approval:** Routing orders to finance or management for manual review and approval based on predefined rules (e.g., order value, customer credit history).
 - **Status Tracking:** Clear visibility of the credit approval status for each order.
3. **Order Fulfillment and Logistics:** This component will manage the delivery of trusses and sundry items to the site.
- **Picking and Packing:** Generating picking lists for warehouse staff and managing packing processes.
 - **Shipping and Delivery Scheduling:** Integrating with logistics providers or internal fleet management for scheduling deliveries. This could include optimizing delivery routes and tracking delivery status.
 - **Proof of Delivery:** Capturing proof of delivery (e.g., digital signatures, photos) upon successful delivery.
4. **Installation Control:** For on-site installation, the module will provide functionalities to:
- **Schedule Installation Teams:** Assigning and scheduling internal or external installation teams.
 - **Progress Tracking:** Monitoring the progress of installation activities.
 - **Issue Management:** Recording and managing any issues or delays encountered during installation.
 - **Completion Sign-off:** Obtaining sign-off upon successful completion of installation.
5. **Integration with Financial Module:** Sales orders will trigger financial entries, including:
- **Invoicing:** Generating invoices based on delivery or completion milestones.
 - **Accounts Receivable:** Managing customer payments and outstanding balances.

- **Revenue Recognition:** Ensuring proper revenue recognition based on accounting standards.

Financial Management Module

Building on the research findings, the Financial Management module will provide comprehensive accounting functionalities, ensuring accurate financial record-keeping and reporting.

Key Features and Data Flow:

1. General Ledger (GL):

- **Centralized Accounting:** The GL will serve as the central repository for all financial transactions, integrating data from Sales (invoices, payments), Purchasing (purchase orders, vendor invoices), Inventory (cost of goods sold, inventory valuation), and Production (work-in-progress, labor costs).
- **Chart of Accounts:** A flexible and customizable chart of accounts to categorize all financial transactions.
- **Automated Journal Entries:** Automatic creation of journal entries from operational transactions (e.g., sales, purchases, inventory movements, production completions).
- **Financial Reporting:** Generation of standard financial statements (Income Statement, Balance Sheet, Cash Flow Statement) and customizable reports for financial analysis.

2. Accounts Receivable (AR):

- **Customer Invoicing:** Generating and sending invoices to customers based on sales orders and delivery/installation milestones.
- **Payment Processing:** Recording customer payments and managing outstanding receivables.
- **Aging Reports:** Providing insights into overdue invoices and facilitating collections.
- **Credit Management:** Integrated with the Order Management module for credit approval and monitoring customer credit limits.

3. Accounts Payable (AP):

- **Vendor Invoice Processing:** Receiving, verifying, and processing vendor invoices against purchase orders.
- **Payment Management:** Scheduling and executing payments to suppliers.
- **Expense Management:** Tracking and categorizing operational expenses.
- **Aging Reports:** Managing outstanding payables and optimizing payment terms.

4. Bank Reconciliation: Automating the process of matching bank statement transactions with internal ledger entries. This will include:

- **Automated Matching Rules:** Configurable rules to automatically match transactions based on amount, date, and reference numbers.
- **Exception Handling:** Tools to easily identify and resolve unmatched transactions.
- **Multi-Bank Support:** Ability to reconcile accounts from multiple banks.

5. Fixed Asset Management: Tracking and managing the company's fixed assets, including depreciation calculations and asset registers.

6. Tax Management: Capabilities to calculate and report various taxes (e.g., sales tax, VAT) in compliance with local regulations.

7. Budgeting and Forecasting: Tools to create and manage budgets, compare actual performance against budgets, and generate financial forecasts.

Integration with Dynamics 365 (Co-existence or Migration)

The strategy for Dynamics 365 integration will depend on the user's long-term vision. Two primary approaches are considered:

1. **Phased Migration:** If the bespoke ERP is intended to fully replace Dynamics 365, a phased migration strategy will be adopted. This involves:
 - **Data Extraction:** Extracting historical data from Dynamics 365 (e.g., customer records, historical quotes, financial transactions) using its APIs or export

functionalities.

- **Data Transformation:** Transforming the extracted data to fit the new ERP's data model.
- **Data Loading:** Loading the transformed data into the new ERP system.
- **Validation and Reconciliation:** Rigorous testing and reconciliation to ensure data accuracy and completeness in the new system.
- **Cutover Strategy:** A carefully planned cutover to transition users from Dynamics 365 to the new ERP system for specific functionalities.

2. **Co-existence and Data Synchronization:** If Dynamics 365 will continue to be used for certain functions (e.g., CRM, specific financial reporting not yet replicated in the bespoke ERP), a co-existence strategy will be implemented. This involves:

- **API-based Integration:** Utilizing Dynamics 365 APIs to synchronize data in real-time or near real-time between the two systems. This could involve pushing new customer data from the bespoke ERP to Dynamics 365, or pulling updated financial information from Dynamics 365 into the bespoke ERP.
- **Middleware:** Leveraging Azure Integration Services (e.g., Azure Logic Apps, Azure Service Bus) to orchestrate data flows and transformations between the bespoke ERP and Dynamics 365, ensuring data consistency across both platforms.
- **Clear Data Ownership:** Defining which system is the

single source of truth for specific data entities to avoid conflicts and ensure data integrity.

Regardless of the chosen approach, the integration will aim to minimize manual data entry and duplication, ensuring a seamless flow of information across the enterprise.

Technology Stack

The existing development using .NET and C# hosted on Microsoft Azure provides a strong foundation. The proposed architecture will leverage this, along with other Azure services, to build a scalable, secure, and performant ERP system.

- **Backend:** .NET Core (C#) for API development and business logic.

- **Database:** Azure SQL Database for relational data storage, offering scalability, high availability, and robust security features. Consideration for Azure Cosmos DB for non-relational data if needed (e.g., for document storage or audit logs).
- **Frontend:** A modern JavaScript framework (e.g., React, Angular, or Vue.js) for building a responsive and intuitive web-based user interface. This aligns with the developer's use of .NET, as Blazor (a .NET web framework) could also be considered for full-stack .NET development.
- **Cloud Platform:** Microsoft Azure, utilizing services such as:
 - **Azure App Service:** For hosting web applications and APIs.
 - **Azure Functions/Logic Apps:** For serverless execution of background processes, data transformations, and integration workflows (e.g., processing Pamir export files, triggering notifications).
 - **Azure Blob Storage:** For storing unstructured data, such as Pamir export files, generated PDF quotations, and other documents.
 - **Azure Key Vault:** For securely managing application secrets, API keys, and connection strings.
 - **Azure Monitor & Application Insights:** For monitoring application performance, logging, and diagnostics.
 - **Azure DevOps:** For continuous integration and continuous delivery (CI/CD) pipelines, enabling automated builds, testing, and deployments.
- **Integration:** Azure Integration Services (Logic Apps, Service Bus, Event Grid) for orchestrating data flows and connecting with external systems like Dynamics 365.

Security Considerations

Security will be a paramount concern throughout the design and implementation of the ERP system. Key security measures will include:

1. **Authentication and Authorization:** Implementing robust identity management using Azure Active Directory (AAD) for user authentication and role-based access control

(RBAC) to ensure users only have access to the data and functionalities relevant to their roles.

2. **Data Encryption:** Encrypting data at rest (in Azure SQL Database and Blob Storage) and in transit (using HTTPS/TLS for all communication).
3. **Network Security:** Implementing Azure Virtual Networks (VNets), Network Security Groups (NSGs), and Azure Firewall to create secure network boundaries and control traffic flow.
4. **Application Security:** Following secure coding practices, regularly performing security audits, and protecting against common web vulnerabilities (e.g., SQL injection, cross-site scripting).
5. **Auditing and Logging:** Comprehensive logging of all system activities and user actions for auditing purposes and to detect suspicious behavior.
6. **Regular Backups and Disaster Recovery:** Implementing automated backup strategies for all data and establishing a disaster recovery plan to ensure business continuity.
7. **Compliance:** Ensuring the system design and data handling practices comply with relevant industry regulations and data privacy laws.

By adhering to these security principles, the bespoke ERP system will provide a secure and reliable platform for managing critical business operations.

Implementation Roadmap and Recommendations

Building a comprehensive ERP system is a significant undertaking that requires a phased approach to ensure successful delivery, manage complexity, and provide early value. This roadmap outlines a recommended sequence of development phases, prioritizing core functionalities and critical integrations. Each phase builds upon the previous one, allowing for iterative development, testing, and feedback.

Phased Implementation Roadmap

Phase 1: Core Quotation and Stock Management (Focus: Pamir Integration & Quoting)

- **Objective:** Establish the foundation for dynamic quotation generation by integrating with Mitek Pamir exports and building a robust stock management module for timber products.
- **Key Activities:**
 - **Detailed Pamir Export Analysis:** In-depth analysis of actual Pamir Excel/JSON export files to precisely identify all relevant variables, their data types, and relationships. This will involve working closely with the user to understand the nuances of their Pamir output.
 - **Pamir Data Ingestion Engine Development:** Develop the .NET/C# parsing engine to read, extract, validate, and transform data from Pamir exports into the ERP's data model. This engine should be resilient to minor variations in export formats.
 - **Core Stock Management Module Development:** Implement the enhanced `StockItems` module with granular attributes for timber (dimensions, grade, species, treatment), flexible UOM conversions, and real-time inventory tracking for raw materials.
 - **Quotation Management Module (MVP):** Develop the initial version of the quotation module, focusing on:
 - Dynamic `QuoteLineItem` generation based on Pamir-extracted variables and predefined formulas.
 - A formula engine to calculate quantities and costs for stock items.
 - Integration with the new Stock Management module for real-time availability checks.
 - Basic quote creation, saving, and PDF export functionality.
 - **User Interface Development:** Create intuitive UI for Pamir file upload, stock item management, and quote creation/editing.
 - **Deliverables:** Functional Pamir data ingestion, core stock management, and MVP quotation module. Ability to generate accurate quotes from Pamir exports.

- **Success Metrics:** Accuracy of Pamir data extraction, successful generation of quotations with correct quantities and pricing, positive user feedback on the quoting process.

Phase 2: Advanced Quotation, BOM, and Order Management (Focus: Production & Sales Workflow)

- **Objective:** Extend quotation capabilities, implement dynamic BOM generation for trusses, and establish the core order management workflow.
- **Key Activities:**
 - **Advanced Quotation Features:** Implement interactive quote line item grid with real-time filtering, template-based quoting, and robust revision/option management (`QuoteVersions` , `QuoteHistories`).
 - **Dynamic BOM & Assembly Management:** Develop the dynamic BOM generation logic for roof trusses based on Pamir data, including parametric BOMs and rules-based configuration. Implement the management of fixed assemblies for tenders.
 - **Production Order Creation:** Enable the automatic generation of production orders from approved quotations, linking to the dynamically created BOMs.
 - **Core Order Management Module:** Develop functionalities for converting approved quotes to sales orders, basic order tracking, and integration with the credit approval workflow (initial manual process).
 - **Credit Approval Workflow (MVP):** Implement the initial credit approval process, potentially starting with manual approvals and status tracking.
- **Deliverables:** Enhanced quotation module, functional dynamic BOM generation, initial production order creation, and core sales order management.
- **Success Metrics:** Streamlined quoting process, accurate BOM generation, efficient conversion of quotes to orders.

Phase 3: Financial Integration and Production Control (Focus: End-to-End Financials & Operations)

- **Objective:** Integrate core financial modules and enhance production control capabilities to provide a comprehensive view of business operations.
- **Key Activities:**
 - **General Ledger (GL) Implementation:** Set up the GL with a flexible chart of accounts. Implement automated journal entries from sales, purchases, and inventory movements.
 - **Accounts Receivable (AR) & Accounts Payable (AP):** Develop modules for customer invoicing, payment processing, vendor invoice processing, and payment management.
 - **Bank Reconciliation:** Implement automated bank reconciliation features with configurable matching rules.
 - **Production Control Enhancements:** Develop work order management, scheduling integration, material issuance tracking, and progress tracking for production orders.
 - **Supplier Management:** Centralized management of supplier information and purchase order tracking.
- **Deliverables:** Fully integrated financial modules (GL, AR, AP, Bank Recon), enhanced production control, and comprehensive supplier management.
- **Success Metrics:** Accurate financial reporting, efficient bank reconciliation, improved control over purchasing and production processes.

Phase 4: Logistics, Installation, and Advanced Features (Focus: Supply Chain & Optimization)

- **Objective:** Complete the end-to-end ERP functionality by adding logistics, installation control, and advanced features for optimization.
- **Key Activities:**
 - **Order Fulfillment & Logistics:** Implement picking/packing, shipping/delivery scheduling, and proof of delivery functionalities.

- **Installation Control:** Develop features for scheduling installation teams, tracking progress, and managing on-site issues.
- **Advanced Inventory Features:** Implement batch/lot tracking, demand forecasting, and reorder point management.
- **Reporting & Analytics Enhancements:** Develop advanced dashboards and custom reports for business intelligence across all modules.
- **Dynamics 365 Integration (Final Phase):** Based on the chosen strategy (migration or co-existence), implement the full integration with Dynamics 365 for historical data migration or ongoing data synchronization.
- **Deliverables:** Complete ERP system with all requested functionalities, optimized supply chain processes, and comprehensive reporting.
- **Success Metrics:** Seamless end-to-end workflow, improved efficiency in logistics and installation, data-driven decision making.

General Recommendations for Success

1. **Iterative Development and Agile Methodologies:** Given the complexity and bespoke nature of the ERP, adopt an agile development approach. This allows for continuous feedback, flexibility to adapt to changing requirements, and faster delivery of working software. Break down each phase into smaller, manageable sprints.
2. **User Involvement and Feedback:** Involve key business users from all departments (design, sales, production, finance, installation) throughout the development lifecycle. Regular demonstrations and feedback sessions are crucial to ensure the system meets their needs and to foster user adoption.
3. **Data Quality and Migration Strategy:** Prioritize data quality from the outset. Develop a clear data migration strategy for existing data (from Pamir, Dynamics 365, and other sources). Invest time in data cleansing and validation to ensure the new system operates with accurate information.
4. **Comprehensive Testing:** Implement a rigorous testing strategy, including unit testing, integration testing, system testing, and user acceptance testing (UAT). Pay special

attention to the Pamir data ingestion and quotation calculation logic, as these are critical to the system's core functionality.

5. **Documentation:** Maintain thorough documentation for all aspects of the system, including technical architecture, data models, business logic, and user manuals. This is essential for ongoing maintenance, future enhancements, and training new users.
6. **Scalability and Performance Considerations:** Design the system with scalability in mind from the beginning. Leverage Azure's elastic capabilities to ensure the system can handle increasing data volumes and user loads. Regularly monitor performance and optimize as needed.
7. **Security Best Practices:** Continuously adhere to the security considerations outlined in the architecture section. Regular security audits and penetration testing should be part of the development and deployment process.
8. **Training and Change Management:** Develop a comprehensive training program for all users. A well-executed change management strategy will help users adapt to the new system and embrace the new workflows, minimizing resistance and maximizing adoption.
9. **Phased Rollout:** Consider a phased rollout of the system, starting with a pilot group or a specific module, before a full company-wide deployment. This allows for lessons learned and adjustments before wider implementation.
10. **Vendor Support and Community Engagement:** For technologies like .NET, C#, Azure, and Dynamics 365, leverage Microsoft's extensive documentation, support resources, and community forums for troubleshooting and best practices.

By following this roadmap and adhering to these recommendations, the bespoke ERP system can be successfully developed and implemented, providing significant operational efficiencies and strategic advantages for the timber roof business.

References

[1] Microsoft Learn. (2024, February 1). *Choose the right pattern for your integration strategy - Dynamics 365*. Retrieved from <https://learn.microsoft.com/en->

us/dynamics365/guidance/implementation-guide/integrate-other-solutions-choose-pattern

[2] Oracle. (2024, May 31). *What is an ERP Finance Module? Features and Benefits*. Retrieved from <https://www.oracle.com/erp/erp-finance-module/>

[3] ECI Solutions. (2023, November 30). *BOM Management and The Product Configurator*. Retrieved from <https://www.ecisolutions.com/blog/manufacturing/deacom-erp-software/bom-management-and-the-product-configurator/>

[4] MRPeasy. (2024, October 22). *Configurable BOM (CBOM) – Simplifying the Production ...*. Retrieved from <https://www.mrpeasy.com/blog/configurable-bom-cbom/>

[5] Intact Software. (2024, January 23). *A Quick Guide to Essential ERP Modules for Lumber Dealers*. Retrieved from <https://www.intactsoftware.com/en-us/genetiq-blog/essential-erp-modules-for-lumber-dealers-a-quick-guide>

[6] Timbersoft. *Inventory Management for Wood Industries*. Retrieved from <https://www.timbersoft.com/>

[7] Autodesk Construction Cloud. (2023, August 15). *Best Practices to Successfully Integrate your ERP and Cost ...*. Retrieved from <https://construction.autodesk.com/resources/source-acc-webinars/best-practices-financial-system-integration/>

[8] Trimble. (2019, September 27). *How BIM and ERP Work Together for Better Precast Workflows*. Retrieved from <https://www.trimble.com/blog/construction/en-US/article/how-bim-and-erp-work-together-for-better-precast-workflows>

[9] CTND. (2024, November 19). *5 Best Practices in ERP System Integrations*. Retrieved from <https://www.ctnd.com/5-best-practices-in-erp-system-integrations/>

[10] Technostruct Academy. (2025, January 8). *A Complete Guide on Construction ERP and BIM Integration*. Retrieved from <https://www.technostructacademy.com/blog/construction-erp-and-bim-integration/>

[11] OpenBOM. (2024, March 2). *CAD to ERP Integration Controversy: Bridging the Engineering To ...*. Retrieved from <https://www.openbom.com/blog/cad-to-erp-integration-controversy-bridging-the-engineering-to-manufacturing-gap>

- [12] The Access Group. (2025, March 14). *Best Practices for Construction Data Management: Key takeaways* Retrieved from <https://www.theaccessgroup.com/en-us/construction/resources/construction-data-management/>
- [13] Trimble. (2021, September 14). *7 Best Practices for Using a Common Data Environment on* Retrieved from <https://www.trimble.com/blog/construction/en-US/article/7-best-practices-for-using-a-common-data-environment-on-complex-construction-projects>
- [14] ResearchGate. (2019, August 26). *(PDF) Sustainability via ERP and BIM Integration*. Retrieved from https://www.researchgate.net/publication/335023856_Sustainability_via_ERP_and_BIM_Integration
- [15] Caddi. *How Integrating Drawing Management with ERP Boosts Efficiency*. Retrieved from <https://us.caddi.com/resources/insights/drawing-management-with-erp>
- [16] Abas. *How does ERP/CAD integration help custom manufacturers?*. Retrieved from <https://abas-erp.com/en/blog/how-does-erpcad-integration-help-custom-manufacturers>
- [17] BigCommerce. *ERP Integrations: Common Patterns and Methods + Best Practices*. Retrieved from <https://www.bigcommerce.com/articles/ecommerce/erp-integration/>
- [18] NetSuite. *ERP Integration: Strategy, Benefits and Best Practices*. Retrieved from <https://www.netsuite.com/portal/resource/articles/erp/erp-integration-strategy.shtml>
- [19] LinkedIn. (2023, June 15). *Best Practices for ERP Integration Architecture and Design*. Retrieved from <https://www.linkedin.com/advice/0/how-do-you-document-maintain-your-erp>
- [20] Godlan. *CAD Integration & ERP Systems: Are You Set Up for Success?*. Retrieved from <https://godlan.com/cad-erp-integration/>
- [21] Elmo Solutions. *(PDF) a new approach to cad/erp data integration*. Retrieved from https://elmosolutions.com/wp-content/uploads/2015/11/White_Paper_Elmo_CAD-ERP_Connectivity-EN.pdf
- [22] Aicomp. (2023, May 23). *Synchronize your CAD workflow with your ERP*. Retrieved from <https://www.aicomp.com/news/press/synchronize-your-cad-workflow-with-your-erp/>

- [23] Top10ERP.org. *The Top ERP Systems for the Lumber and Wood Industries*. Retrieved from <https://www.top10erp.org/erp-software-comparison/best-fit/lumber-and-wood>
- [24] acadon AG. *ERP Software for timber trade*. Retrieved from <https://acadon.net/en/erp-software/timber-trade/>
- [25] Friedman Corp. (2024, June 27). *Maximize Efficiency with Woodworking Design Software and ERP*. Retrieved from <https://friedmancorp.com/blog/woodworking-design-software-and-erp/>
- [26] VlexPlus. *ERP software for wood, wood processing & the timber industry*. Retrieved from <https://www.vlexplus.com/en/industries/erp-system-for-wood-wood-industry-wood-processing>
- [27] ECI Solutions. *Spruce: ERP and POS for Lumber & Building Materials Dealers*. Retrieved from <https://www.ecisolutions.com/products/building-materials-software/>
- [28] ERP Absolute. *ERP Software for Wood Manufacturing Industry*. Retrieved from <https://www.erpabsolute.com/manufacturing-erp/wood-industry-erp/>
- [29] DMSi. *DMSi Agility*. Retrieved from <https://www.dmsi.com/products/agility-erp/>
- [30] TIMMSanywhere. *Home / TIMMSanywhere - Modern Timber Software Solution*. Retrieved from <https://www.timmsanywhere.com/>
- [31] acadon AG. (2025, February 26). *Timber Solutions Software: 5 Systems to Boost your Business*. Retrieved from <https://acadon.net/en/blog/timber-solutions-software/>
- [32] Timbersoft. *Lumber Mill Inventory Management*. Retrieved from <https://www.timbersoft.com/lumber-mill-inventory-management>
- [33] Virginia Tech. (2021, May 3). *Lean Inventory Management in the Wood Products Industry*. Retrieved from <https://www.pubs.ext.vt.edu/420/420-148/420-148.html>
- [34] 3log.com. *Log Inventory & Management System (LIMS)*. Retrieved from <https://www.3log.com/products/lims-log-inventory-management-system/>
- [35] BPI. (2024, September 18). *Lumber Inventory Optimization & Backorder Best Practices*. Retrieved from <https://marketing.bpi.build/inventory-optimization/>

- [36] ECI Solutions. *Software for Lumber Yards*. Retrieved from <https://www.ecisolutions.com/industries/building-supply/lumberyards/>
- [37] Buske. *Lumber and Timber 3PL Fulfillment and Warehousing*. Retrieved from <https://www.buske.com/industries-served/lumber-timber>
- [38] Milltech IMS. (2024, July 30). *A Lumber Inventory Management System Software's 4 most* Retrieved from <https://www.milltechims.com/post/a-lumber-inventory-management-software-s-4-most-important-features>
- [39] WoodPro Software. *Lumber Inventory Software*. Retrieved from <https://www.woodprosoftware.com/distribution.html>
- [40] Reddit. (2023, July 18). *How Do You Inventory?*. Retrieved from https://www.reddit.com/r/woodworking/comments/1531vfj/how_do_you_inventory/
- [41] Acctivate. *Lumber Inventory Software*. Retrieved from <https://acctivate.com/industries/lumber-inventory-software/>
- [42] Craftybase. *Woodworking Inventory Software*. Retrieved from <https://craftybase.com/woodworking-inventory-software>
- [43] Intact Software. (2024, October 21). *Best Lumber ERP Software: 8 Key Features for Your Business*. Retrieved from <https://www.intactsoftware.com/en-us/genetiq-blog/best-lumber-erp-software-must-have-features>
- [44] Manufacturing Tomorrow. (2024, November 11). *Configurable BOMs – Simplifying the Production of Customizable* Retrieved from <https://www.manufacturingtomorrow.com/article/2024/11/configurable-boms-%E2%80%93-simplifying-the-production-of-customizable-items/23804>
- [45] ECI Software Solutions. *Product Configurator in M1 ERP*. Retrieved from <https://www.ecisolutions.com/products/erp-software-for-manufacturing/features/production-management/product-configurator/>
- [46] Steersman. (2023, December 15). *ERP and CPQ with dynamic BOMs for make-to-order manufacturers*. Retrieved from <https://steersman.works/a/94-erp-with-dynamic-bills-of-material-for-semi-custom-manufacturers>

- [47] OmegaCube Technologies. *Bill of Materials (BOM)*. Retrieved from <https://www.omegacube.com/p/erp/engineering/bill-of-materials-bom/>
- [48] SAP Help Portal. *Configurable Materials*. Retrieved from https://help.sap.com/docs/SAP_ERP/0182e2bf25e64b2b9743e81300cc9764/e2078ca4a1e5409cb2cdec249fbcc828.html?locale=en-US&state=PRODUCTION&version=6.05.latest
- [49] DealHub. (2025, May 16). *What is an ERP Product Configurator?*. Retrieved from <https://dealhub.io/glossary/erp-product-configurator/>
- [50] Friedman Corp. *Powerful Parametric Product Configurator By Frontier ERP*. Retrieved from <https://friedmancorp.com/parametric-product-configurator/>
- [51] EM Engineering Solutions. (2024, November 10). *Effective Management of BOMs for Product Success*. Retrieved from <https://emengineeringsolutions.com/articles/f/effective-management-of-boms-for-product-success?blogcategory=Research+Studies>
- [52] Manufacturing Tomorrow. (2025, March 18). *BOM Management Expert Guide for Growing Manufacturers*. Retrieved from <https://www.manufacturingtomorrow.com/article/2025/02/bom-management-expert-guide-for-growing-manufacturers/24267/>
- [53] MRPeasy. (2024, December 3). *BOM Management Guide for Growing Manufacturers*. Retrieved from <https://www.mrpeasy.com/blog/bom-management/>
- [54] NetSuite. (2025, June 17). *What Is a Bill of Materials (BOM)? The Complete Guide to BOMs*. Retrieved from <https://www.netsuite.com/portal/resource/articles/erp/bill-of-materials-bom.shtml>
- [55] Gravity Software. *Bill of Materials (BOM) for Manufacturing*. Retrieved from <https://www.gogravity.com/product/operations/bill-of-materials>
- [56] Panorama Consulting. (2020, January 13). *Is ERP Good Enough To Manage Bill Of Materials?*. Retrieved from <https://www.panorama-consulting.com/is-erp-good-enough-to-manage-bill-of-materials/>
- [57] Microsoft Learn. (2024, April 19). *Bills of materials and formulas - Dynamics 365*. Retrieved from <https://learn.microsoft.com/en-us/dynamics365/supply-chain/production-control/bill-of-material-bom>

[58] Dynamics 365 Community. (2024, February 27). *Optional Items in BOMs/Assembly items with variable make-ups*. Retrieved from <https://community.dynamics.com/forums/thread/details/?threadid=f0ea909c-97d5-ee11-9079-00224827e012>

[59] SAP Help Portal. *BOM Usage*. Retrieved from https://help.sap.com/docs/SAP_ERP/43fa3cddb00348beb3ed12212a94448f/3c05c453f57eb44ce10000000a174cb4.html

[60] Reddit. (2024, January 5). *How are folks doing BOM management for new product designs?*. Retrieved from https://www.reddit.com/r/MechanicalEngineering/comments/18z0oqi/how_are_folks_doing_bom_management_for_new/

[61] Prometheus Group. *Bill of Materials (BOM) Guide*. Retrieved from <https://www.prometheusgroup.com/learning-center/what-is-a-bill-of-materials-bom>

[62] SYSPRO US Blog. (2022, September 20). *Best use of the Bill of Materials (BOM) in ERP*. Retrieved from <https://us.syspro.com/blog/erp-implementation/how-to-make-best-use-of-the-bill-of-materials-bom-in-your-erp/>

[63] ERP Software Blog. (2025, January 17). *Creating and Managing a Comprehensive Bill of Material (BOM) with Dynamics 365*. Retrieved from <https://erpsoftwareblog.com/2025/01/creating-and-managing-a-comprehensive-bill-of-material-with-dynamics-365/>

[64] Acumatica Community. (2024, September 4). *Dynamic BOM calculations*. Retrieved from <https://community.acumatica.com/low-code-no-code-customizations-or-integrations-291/dynamic-bom-calculations-26149>

[65] Messe.de. (PDF) *DYNAMIC BILL OF MATERIALS*. Retrieved from https://www.messe.de/apollo/hannover_messe_2020/obs/Binary/A1003793/1003793_02207086.pdf

[66] Datix. *Simplify Bill of Materials with ERP*. Retrieved from <https://datixinc.com/blog/simplify-bill-of-materials-with-erp/>

- [67] Microsoft Learn. (2024, June 11). *Work with bills of material - Business Central*. Retrieved from <https://learn.microsoft.com/en-us/dynamics365/business-central/inventory-how-work-boms>
- [68] Total ETO. *Building an ETO BOM - Engineer To Order*. Retrieved from <https://totaleto.com/white-papers/building-an-eto-bom/>
- [69] MRPeasy. (2023, August 28). *Bill of Materials (BOM) – A Complete Guide with Examples*. Retrieved from <https://www.mrpeasy.com/blog/bill-of-materials/>
- [70] Genius ERP. *How to Make a Bill of Materials (BOM) the Right Way*. Retrieved from <https://www.geniuserp.com/resources/blog/how-to-make-a-bill-of-materials-the-right-way/>
- [71] Panni Management. *Bill of Materials Explained | What is BOM in ERP Software*. Retrieved from <https://pannimanagement.com/blog/bom-in-erp-software-explained/>
- [72] The CFO Club. (2025, May 28). *24 Best ERP Systems For Custom Manufacturing Reviewed In 2025*. Retrieved from <https://thecfoclub.com/tools/best-erp-for-custom-manufacturing/>
- [73] SYSPRO. *BOM Bill of Materials | ERP for Manufacturers*. Retrieved from <https://us.syspro.com/business-software/business-management-software/manufacturing-bill-of-materials-bom/>
- [74] Bommer. (2024, September 23). *The Critical Role of the Bill of Materials in ERP Systems*. Retrieved from <https://www.getbommer.com/post/role-of-bom-in-erp/>
- [75] QT9 ERP. (2025, February 3). *Supercharge Manufacturing Operations with Bill of Materials Software*. Retrieved from <https://qt9erp.com/blog/bill-of-materials-software-bom>
- [76] Elmo Solutions. *What Do We Need to Know About Bills of Materials (BoM) in an ERP* Retrieved from <https://elmosolutions.com/boms-erp/>
- [77] NetSuite. (2024, August 16). *What Is an ERP Financial Module? Key Components and Benefits*. Retrieved from <https://www.netsuite.com/portal/resource/articles/erp/erp-finance-module.shtml>

- [78] ERP Focus. (2024, November 8). *ERP finance modules guide (finance features in ERP)*. Retrieved from <https://www.erpfocus.com/erp-finance-module-features.html>
- [79] Tyler Technologies. *ERP Pro Bank Reconciliation*. Retrieved from <https://www.tylertech.com/resources/resource-downloads/erp-pro-bank-reconciliation>
- [80] Unit4. *Unit4 financial management ERP software module*. Retrieved from <https://www.unit4.com/products/erp-accounting-software/financial-management>
- [81] Zone & Co. (2025, June 23). *Automating bank reconciliation in ERP*. Retrieved from <https://www.zoneandco.com/articles/finance-teams-guide-to-erp-bank-reconciliation-automation-challenges-best-practices-and-proven-benefits>
- [82] TechTarget. (2023, November 27). *What is an ERP Finance Module? | Definition from TechTarget*. Retrieved from <https://www.techtarget.com/searcherp/definition/ERP-finance-module>
- [83] VersaCloud ERP. (2024, November 6). *Essentials of Accounting in an ERP: What is Bank Reconciliation*. Retrieved from <https://www.versaclouderp.com/blog/essentials-of-accounting-in-an-erp-what-is-bank-reconciliation/>
- [84] Top10ERP.org. *ECi M1 Financial Management Modules Overview*. Retrieved from <https://www.top10erp.org/products/eci-m1/financial-management>
- [85] SAP B1 Blog. (2025, January 17). *The finance module - the heart of the ERP software*. Retrieved from <https://sap-b1-blog.com/en/the-financial-module-at-the-heart-of-the-erp-software/>
- [86] ControlHub. (2024, March 21). *General Ledger Best Practices: Ensuring Financial Success*. Retrieved from <https://www.controlhub.com/blog/gl-reconciliation-best-practices>
- [87] Syspro. (2022, December 13). *Getting value from the general ledger within your ERP system*. Retrieved from <https://www.syspro.com/blog/erp-for-finance/getting-value-from-the-general-ledger-within-your-erp-system/>
- [88] Vareto. (2023, June 15). *Successful ERP implementation: Best practices for data cleanliness* Retrieved from <https://www.vareto.com/blog/buy-erp-software-successful-implementation>

- [89] VNMT Solutions. (2025, April 4). *NetSuite General Ledger Best Practices*. Retrieved from <https://www.vnmtsolutions.com/best-practices-for-optimizing-your-general-ledger-in-netsuite/>
- [90] Advantive. *Best Practices*. Retrieved from <https://advantive.my.site.com/support/s/topic/0TO8b0000006WQEGA2/best-practices>
- [91] Innovature Inc. (2024, September 18). *Top 5 Best Practices for Maintaining an Accurate General Ledger*. Retrieved from <https://innovatureinc.com/top-best-practices-for-general-ledger/>
- [92] Surety Systems. (2024, September 5). *JD Edwards General Ledger: Best Practices for Financial Management*. Retrieved from <https://www.suretysystems.com/insights/jd-edwards-general-ledger-best-practices-for-financial-management/>
- [93] Panorama Consulting. (2024, October 9). *Best Practices For Implementing A Finance ERP System*. Retrieved from <https://www.panorama-consulting.com/best-practices-for-implementing-a-finance-erp-system/>
- [94] Caisoft. *The Benefits of ERP Systems for General Ledger Management*. Retrieved from <https://caisoft.com/resources/general-ledger-module-in-erp/>
- [95] Oracle Blogs. (2025, May 30). *How to Secure Access to Data in General Ledger*. Retrieved from <https://blogs.oracle.com/erp-ace/post/how-to-secure-access-to-data-in-general-ledger>
- [96] Microsoft Learn. *Reconcile bank statements by using advanced bank reconciliation*. Retrieved from <https://learn.microsoft.com/en-us/dynamics365/finance/cash-bank-management/reconcile-bank-statements-advanced-bank-reconciliation>
- [97] HighRadius. *Bank Reconciliation Software - Simplify Workflows with Automation*. Retrieved from <https://www.highradius.com/product/bank-reconciliation-software/>
- [98] Deskera. *Bank Reconciliation*. Retrieved from <https://www.deskera.com/accounting/bank-reconciliation>
- [99] Nomentia. (2025, March 28). *Best 10 bank reconciliation software*. Retrieved from <https://www.nomentia.com/blog/best-10-bank-reconciliation-software>

- [100] Paystand. (2024, August 23). *Bank Reconciliation Automation: FAQs, Benefits, and Tools*. Retrieved from <https://www.paystand.com/blog/bank-reconciliation-automation>
- [101] Sage US. *Bank reconciliation software*. Retrieved from <https://www.sage.com/en-us/integrated-payment-solutions/bank-reconciliation-software/>
- [102] Net at Work. (PDF) *Sage 100 ERP*. Retrieved from https://www.netatwork.com/uploads/pdfs/sage_100_bank_reconciliation.pdf
- [103] Tipalti. *Implement a Purchase Order Process in 7 Steps*. Retrieved from <https://tipalti.com/resources/learn/implementing-purchase-order-process-erp/>
- [104] Acumatica. *Purchase Order Management - Cloud ERP*. Retrieved from <https://www.acumatica.com/cloud-erp-software/distribution-management/purchase-order-management/>
- [105] Spendflo. (2025, February 28). *5 Best Purchase Order Management Softwares for 2025*. Retrieved from <https://www.spendflo.com/blog/top-purchase-order-management-software>
- [106] SoftEngine. (2023, October 11). *Purchase Order Management with ERP: Streamline Your Business*. Retrieved from <https://softengine.com/purchase-order-management-with-erp-streamline-your-business/>
- [107] SourceDay. (2025, January 28). *Mastering Purchase Order Management: Strategies for Efficiency* Retrieved from <https://sourceday.com/blog/purchase-order-management/>
- [108] VersaCloud ERP. (2024, September 6). *ERP Purchase Order Automation: Key Advantages for Businesses*. Retrieved from <https://www.versaclouderp.com/blog/erp-purchase-order-automation-key-advantages-for-businesses/>
- [109] Genius ERP. *How to Use an ERP to Improve your Purchasing Processes*. Retrieved from <https://www.geniuserp.com/resources/blog/how-to-use-an-erp-to-improve-your-purchasing-processes/>
- [110] Tipalti. *End-to-End Purchase Order Management Software*. Retrieved from <https://tipalti.com/procurement/po-management/>
- [111] i95Dev. *Transform Procurement with ERP-Integrated Purchase Order Systems*. Retrieved from <https://www.i95dev.com/use-cases/transform-procurement-with-erp->

[integrated-purchase-order-systems/](#)

[112] SAP Learning. *Executing Purchase Order Management*. Retrieved from <https://learning.sap.com/learning-journeys/executing-basic-erp-processes-with-sap-s-4hana/executing-purchase-order-management>

[113] Microsoft Learn. (2022, September 29). *Integration between finance and operations apps and third-party* Retrieved from <https://learn.microsoft.com/en-us/dynamics365/fin-ops-core/dev-itpro/data-entities/integration-overview>

[114] Microsoft Learn. (2022, February 23). *Best Practices: Integration Patterns*. Retrieved from <https://learn.microsoft.com/en-us/shows/dynamics-365-fasttrack-architecture-insights/best-practices-integration-patterns>

[115] Dynamics 365 Community. (2022, March 25). *Integration patterns in D365FO*. Retrieved from <https://community.dynamics.com/blogs/post/?postid=2c03984b-dcb4-44ba-82dd-2eb7e23a715c>

[116] Azure Integrations. (2020, May 3). *Integration Design Patterns for D365 FO*. Retrieved from <https://azureintegrations.com/2020/05/03/dynamics-365-uo-integration-patterns/>

[117] YouTube. (2023, December 22). *Integration Patterns for Dynamics 365 Finance and Operations* Retrieved from <https://www.youtube.com/watch?v=tUMEZIEKjEQ&pp=0gcJCfwAo7VqN5tD>

[118] CRM Software Blog. (2025, March 26). *5 Data Integration Best Practices for Dynamics 365 Sales*. Retrieved from <https://www.crmsoftwareblog.com/2025/03/5-data-integration-best-practices-for-dynamics-365-sales/>

[119] Dynamics 365 Community. (2023, December 22). *Integration Patterns for Dynamics 365 Finance and Operations* Retrieved from <https://community.dynamics.com/blogs/post/?postid=27adb1ef-d19d-ee11-be37-000d3a4e511f>

[120] Stallions Solutions. (2023, June 13). *Unifying Business Processes with Dynamics 365 Integration*. Retrieved from <https://stallions.solutions/unifying-business-processes-with-dynamics-365-integration/>

[121] WaferWire. *Guide to Dynamics 365 F&O Integration Patterns*. Retrieved from <https://waferwire.com/blog/d365-fo-integration-patterns-guide/>

- [122] Microsoft Learn. *Create a data migration strategy for Dynamics 365 solutions*. Retrieved from <https://learn.microsoft.com/en-us/training/modules/data-migration/>
- [123] Microsoft Learn. (2024, January 23). *Manage configuration and migration data for Dynamics 365 projects*. Retrieved from <https://learn.microsoft.com/en-us/dynamics365/guidance/implementation-guide/data-management-configuration-data-migration>
- [124] SingleStone. *Dynamics 365 Data Migration: A Step-by-Step Guide*. Retrieved from <https://www.singlestoneconsulting.com/blog/dynamics-data-migration>
- [125] Staedean. (2024, April 4). *20 Insider Tips for a Successful Data Migration to D365 F&SCM*. Retrieved from <https://staedean.com/data/blog/data-migration-tips-and-tricks>
- [126] Reddit. (2023, March 22). *Data Migration strategy*. Retrieved from https://www.reddit.com/r/Dynamics365/comments/11yqpeb/data_migration_strategy/
- [127] Intellias. (2025, March 20). *Microsoft Dynamics 365 Migration: A Step-By-Step Guide*. Retrieved from <https://intellias.com/microsoft-dynamics-365-migration-guide/>
- [128] Dynamics 365 Community. (2025, May 9). *Data migration strategy from GP to BC*. Retrieved from <https://community.dynamics.com/forums/thread/details/?threadid=76fbc243-e82c-f011-8c4e-0022481fa6ba>
- [129] Reddit. (2025, January 1). *Data Migration - D365 CE (crm)*. Retrieved from https://www.reddit.com/r/Dynamics365/comments/1hrcql9/data_migration_d365_ce_crm/
- [130] CRM Software Blog. (2025, April 8). *Smooth Data Migration to Dynamics 365 ERP / CRM Strategies and Best Practices*. Retrieved from <https://www.crmsoftwareblog.com/2025/04/smooth-data-migration-to-dynamics-365-erp-crm-strategies-and-best-practices/>
- [131] Dynamics 365 Community. (2025, March 26). *Data Migration Strategies for Dynamics 365: A Comprehensive Guide*. Retrieved from <https://community.dynamics.com/blogs/post/?postid=95e2053b-ff0a-f011-bae3-7c1e5266971b>
- [132] Microsoft Dynamics 365. *AI-Powered ERP Solutions*. Retrieved from <https://www.microsoft.com/en-us/dynamics-365/solutions/erp>

- [133] Microsoft Learn. (2024, February 1). *Integrate Dynamics 365 apps with other systems*. Retrieved from <https://learn.microsoft.com/en-us/dynamics365/guidance/implementation-guide/integrate-other-solutions-guidance-product>
- [134] Zealousys. (2024, July 9). *Integrating CRM and ERP With Microsoft Dynamics 365*. Retrieved from <https://www.zealousys.com/blog/integrating-crm-and-erp-with-microsoft-dynamics-365/>
- [135] Autodesk Construction Cloud. *Dynamics 365 Integration*. Retrieved from <https://construction.autodesk.com/workflows/construction-software-integrations/dynamics-365/>
- [136] Paragon. *Build a native Microsoft Dynamics 365 ERP integration*. Retrieved from <https://www.useparagon.com/integrations/microsoft-dynamics-365-erp>
- [137] Aonflow. *Microsoft Dynamics 365 ERP Integration*. Retrieved from <https://www.aonflow.com/microsoft-dynamics-365-integration>
- [138] Osmosys. (2024, July 15). *Integrate D365 Sales Easily with an ERP using Custom Connectors*. Retrieved from <https://osmosys.co/blog/custom-connector-for-d365sales/>
- [139] Brainvire. (2024, October 3). *Integrating CRM and ERP With Microsoft Dynamics 365*. Retrieved from <https://www.brainvire.com/blog/integrating-microsoft-dynamics-365-with-crm-and-erp/>
- [140] CRM Software Blog. (2024, October 30). *Data Integration between Microsoft Dynamics 365 CRM and ERP*. Retrieved from <https://www.crmsoftwareblog.com/2024/10/streamlining-data-integration-between-microsoft-dynamics-365-crm-and-erp/>
- [141] Tipalti. *The Complete Microsoft Dynamics 365 Integration Guide*. Retrieved from <https://tipalti.com/blog/microsoft-dynamics-integration/>